IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 19 February 2025, accepted 9 March 2025, date of publication 18 March 2025, date of current version 26 March 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3552585

RESEARCH ARTICLE

A Flexible Framework for Design and Validation of rPPG Methods

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This work was supported by Singapore Ministry of Education under Grant A-0008302-02-00 and Grant A-8000136-01-00.

ABSTRACT Over the past decade, Remote Photoplethysmography (rPPG) has emerged as an unobtrusive alternative to wearable sensors for measuring physiological signals, such as heart rate. Despite advancements, its real-world scalability remains limited due to the absence of standardized benchmark methods for validation. This lack of standardization complicates proper comparisons between different approaches, creating inconsistencies in performance evaluation. To address this, we conducted a comprehensive review of recent rPPG methods, analyzing their pre- and post-processing algorithms, validation procedures, benchmark algorithms, datasets, evaluation metrics, data segmentation, and reported results. Our findings demonstrate significant variability in the reported Mean Absolute Error (MAE) of benchmark rPPG methods applied to the same public datasets, confirming the challenge of inconsistent evaluation. By examining the original implementations of established benchmark methods, we developed a flexible framework that optimally selects pre- and post-processing algorithms through an exhaustive search. Applying this framework to benchmark algorithms across three public datasets, we found that 80% of the refined methods ranked within the top 25th percentile in MAE, RMSE, and PCC, with 60% surpassing the highest reported accuracies. These refined methods provide a more rigorous foundation for evaluating novel rPPG techniques, addressing the standardization gap in the field. The codebase for this framework (frPPG) is available at [https://github.com/Building-Robotics-Lab/flexible_rPPG], offering a valuable tool for designing and benchmarking rPPG methods against the best-performing algorithms on a given dataset.

INDEX TERMS Benchmark rPPG methods, error variability, heart rate monitoring, open-source code, pre- and post- processing stages and algorithms, remote photoplethysmography (rPPG).

I. INTRODUCTION

Measuring heart rate can provide information about an individual's cardiovascular health and fitness level. Abnormal heart rates, such as a very fast or very slow heart rate, can indicate underlying health problems, such as heart disease or thyroid issues. Heart rate data can also provide important insight in various domains, such as telemedicine, sports, and stress management. Measuring heart rate had traditionally involved invasive contact methods such as the Electrocardiogram (ECG), which tracks electrical signals of the heart using sensors predominantly attached to the chest. While widely available devices like smartwatches utilize Photoplethysmography (PPG) to measure blood volume changes non-invasively, they can still be obtrusive or irritating for some due to skin sensitivity or the disturbance they may cause in certain settings. Moreover, the rise of highly contagious diseases like COVID-19 highlights the critical need for non-contact health monitoring technologies.

Remote Photoplethysmography (rPPG) has emerged as a promising approach to address these challenges, using low-cost cameras to measure heart rate by leveraging the optical properties of skin without physical contact. rPPG methods analyze the light reflected from the skin, typically from a designated facial Region of Interest (ROI), which captures the rhythmic color variations caused by pulsatile changes

The associate editor coordinating the review of this manuscript and approving it for publication was Inês Domingues.

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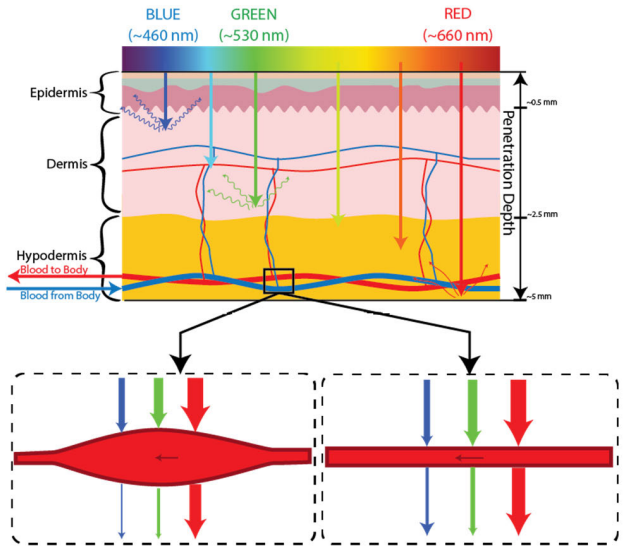


FIGURE 1. Different light absorption across the visible light spectrum due to the heartbeat.

in blood volume and the distinct light absorption behavior of the hemoglobin in the blood. Specifically, the blue light is more strongly absorbed, while green light is moderately absorbed, and red light is minimally absorbed by hemoglobin in the blood [1]. Therefore, when blood volume changes in the cutaneous vessels due to a heartbeat, the absorption pattern of light and thereby the reflected light's spectral density also varies. This results in subtle variations in skin color that are rhythmic and correspond to the heart rate [2] (Figure 1).

Over the past decade, novel rPPG methods in the literature have transitioned from earlier multi-stage unsupervised (heuristic-driven) learning to more recent end-to-end supervised and deep learning (data-driven) algorithms [3], [4]. Multi-stage rPPG methods could be segregated into core rPPG algorithms (e.g., CHROM, POS, ICA) and pre- and post-processing stages (e.g., face detection and tracking, ROI extraction, filtering, signal normalization) which aim to reduce noise and improve signal quality. While supervised methods offer the advantage of potentially handling complex variations in data automatically, their accuracy so far is on par with traditional algorithms [5]. Despite advancements in both unsupervised and supervised rPPG methods, their application in scalable, real-world deployments have encountered significant hurdles which is primarily due to the absence of benchmark methods for validating the novel methods [6], [7], [8], [9]. Without a consensus on benchmarks, evaluations of rPPG methodologies are fragmented and is evident by the high variability of accuracy reported for benchmark methods across public datasets (as shown later in Fig. 4). One of the major reasons for high variability in reported accuracies is the fact researchers often concatenate the core rPPG algorithms with an arbitrary set of pre- and post-processing algorithms to further enhance the results.

While several literature reviews have aimed at establishing and comparing benchmark methods and datasets, they have not systematically examined the literature to identify algorithms used in the pre- and post-processing stages and often selectively pick and describe a handful of them without a reasonable justification [9]. For example, Rouast et al. [9] outlined a general algorithm framework, subdividing typical rPPG algorithms into key components and classifying them accordingly. However, this study did not specifically consider the benchmark algorithms used in each study for the comparison, rather just reported a summary of a few of the published works. Heusch et al. [8] addressed the algorithms in a broad, classificatory sense and focused on three well-known algorithms (e.g., CHROM, LiCVPR, and 2SR) rather than providing detailed, step-by-step algorithmic descriptions. Boccignone et al. [6] introduced one of the most comprehensive framework and codebase for dissecting and studying rPPG algorithms. However, this study did not undertake a systematic process to select pre- and post- processing and core rPPG algorithms and permutating through them, leading to missing out on potentially better performing methods. In addition, they did not explicitly discuss how they established their benchmark algorithms. Selvaraju et al. [10] did a comprehensive review on various aspects of rPPG such as hardware, software, and experimental setup, and provided a detailed summary of pre-processing algorithms, but lacked in post-processing and describing the interdependencies among the algorithms. Premkumar and Hemanth [3] provided a comprehensive overview of the development and methodologies and reviewed a series of articles based on pre- and post-processing algorithms. However, they did not systematically extract the articles and also did not dissect the pre- and post-processing algorithms. Pagano et al. [4] implemented a bibliographic literature review of supervised learning algorithms but did not include any discussion on the performance and comparison of these algorithms for benchmarking. Sikdar et al. [11] highlighted the lack of benchmarking with respect to a challenging dataset, but the algorithm selection did not follow a systematic procedure and algorithms were selected arbitrary. In addition, there is an interdependency among algorithms used in pre- and post-processing stages and the core PPG algorithms [6]. In other words, the best performing set of algorithms could differ if the core rPPG or any pre- or post-processing algorithms is changed [6]. To address this, [7] proposed a framework to permutate through various algorithms in every sub-step of pre- and post-processing to identify the best performing set, but they did not follow a systematic procedure to identify the algorithms and selectively picked a few.

In summary, there is a need for establishing a systematic process for selection of algorithms in pre- and post-processing stages taking into account the interdependency between the algorithms [7]. This necessitates a comprehensive framework for permutating through top-performing algorithms in pre- and post-processing stages for different core rPPG algorithms to identify the best performing architecture of the algorithms. Establishing a public codebase based on this framework is essential for a just benchmarking

and enhancements of the novel algorithms which in turn helps the progression of rPPG from research to reliable real-world applications.

To address the highlighted gaps, we propose a framework that provides a structured classification of existing approaches, highlighting the interdependent impact of pre- and post-processing stages on the accuracy. Through a meticulous literature review, we scrutinize the methodologies, dissect the algorithms and group them into stages, and evaluate the validation practices of existing rPPG research and design a method for identifying the best performing architecture of algorithms. Specifically, we contribute a systematic, data-driven approach to benchmark widely used rPPG algorithms, enabling the identification of optimal algorithmic combinations for different datasets and application scenarios. Moreover, we introduce a flexible search-based mechanism that rapidly explores various algorithmic permutations, thereby uncovering novel insights into how pre- and post-processing choices influence performance. Our study reveals a spectrum of algorithmic strategies, from unsupervised to supervised methods and underscores the transformative role of preprocessing and post-processing in enhancing algorithm performance. This allows researchers to make principled choices about pre- and post-processing methods which could lead to a robust validation protocol that includes a well-defined and high-performing baseline, fostering a proper comparative analysis and set the stage for future innovations that could surmount the current barriers to real-world deployment. In doing so, we not only unify these previously disparate elements into a cohesive and easily replicable framework, but also concretely demonstrate how such systematic selection and permutation of algorithms can significantly elevate the reliability and applicability of rPPG methods. We also describe the open codebase containing the proposed framework and the original implementations of the benchmark algorithms.

The paper is structured as follows: Section II outlines the methodology for literature selection, including the use of the PRISMA workflow, and describes the process of extracting and selecting benchmark methods, datasets, and evaluation metrics, their reported results, and detailing the documentation of pre- and post-processing stages in unsupervised literature, and training and testing metrics in supervised learning literature. Section III presents a comprehensive statistical comparison between the reported results from the literature, original implementations of rPPG methods and their improved versions. Section IV discusses the use cases of the proposed framework, difficulties faced, future work, and privacy concerns. Section V finally draws a conclusion.

II. METHODOLOGY

In this section, we first describe the process of selecting original articles focused on developing novel rPPG methods. We then describe the process of dissecting the articles and extracting pertinent information, such as the algorithms used in pre- and post-processing stages and benchmark algorithms, public datasets, evaluation metrics, and procedure (training and testing data segmentation), reported results. To identify the widely used rPPG methods, public datasets and evaluation metrics, we ranked them based on the number of times each was used in the reviewed articles. To establish the benchmarks, we used the most occurring instance in each category across the studies.

Based on the established benchmarks, we propose a framework for the optimal selection of algorithms in pre- and post-processing stage. In this framework, we specifically focus on the breakdown of the most widely used benchmark rPPG method's stages and the employed algorithms. This framework allows for permutating through the algorithms at different stages to identify the combination that best suits the core rPPG algorithm. Leveraging this flexible framework could lead to improved accuracy through permutations among the top performing algorithm candidates at each stage. We will then evaluate how applying this framework impacts the accuracy of the benchmark rPPG. We finally describe the implementation of the framework into a codebase. The codebase also contains the original implementations of the selected widely used algorithms and are readily compatible with widely used public datasets and offers various evaluation metrics.

A. LITERATURE SELECTION

In order to search through the literature and identify the original articles, we employed the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) workflow [12]. PRISMA is a systematic review methodology that provides a standardized framework for planning, conducting, and reporting systematic reviews and meta-analyses in research. It aims to enhance the transparency and quality of these studies by offering guidelines for each stage, from literature search and selection to data extraction and synthesis, ultimately promoting more rigorous and reproducible research in evidence synthesis. It is widely regarded as the gold standard for systematic reviews due to its rigorous and comprehensive approach. While other standardized workflows such as MOOSE (Meta-analysis Of Observational Studies in Epidemiology), the Cochrane Handbook guidelines, or the Joanna Briggs Institute methodology are available, PRISMA's consistently updated guidelines, extensive community adoption, and clear implementation steps made it the most suitable choice for ensuring methodological soundness and transparency in this review. Scopus, IEEE Xplore, ScienceDirect, and Google Scholar were selected as the literature repositories as they offer an expansive scope of relevant studies. We specifically looked for original articles published after 2005 and used the following keywords: (''remote photoplethysmography'' OR rPPG) AND ''face'' AND ''heart rate'' AND (''public datasets'' OR ''public'').

We then reviewed through the abstract and, if needed, the entire document, and excluded the article if it does not meet any of the following criteria:

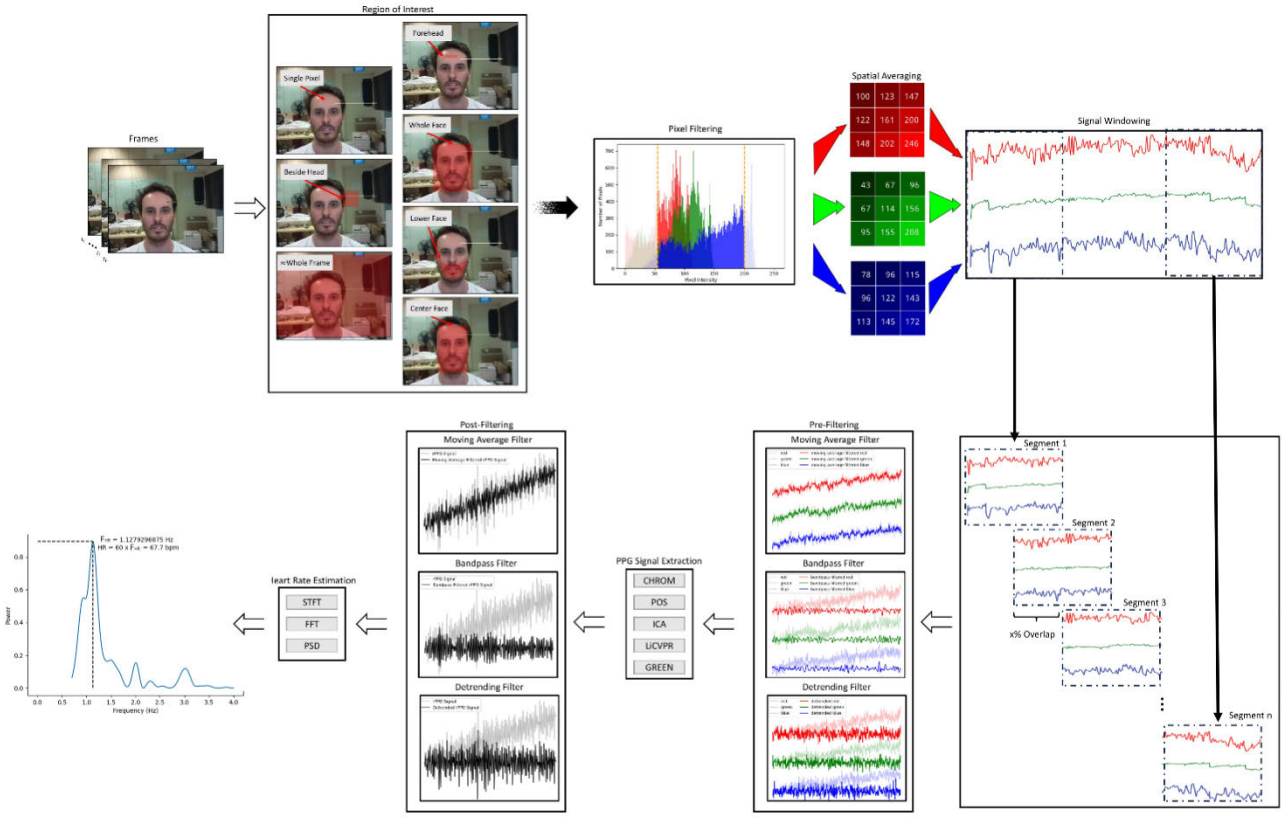


FIGURE 2. **Proposed framework containing the stages in which several algorithms are permuted.**

- The paper should propose a new method and compare it to previously proposed algorithms from rPPG papers.
- There are different areas in the body to which rPPG can be applied. The paper should focus on facial areas.
- Different physiological data can be extracted from the rPPG signal [13]. The paper should have elements focusing on heart rate estimation.
- rPPG researchers either use public datasets or their own private datasets to evaluate their proposed methods. The paper should have included some of the public datasets as part of the validation process.

Additionally, only papers published in English are selected. After using the keywords in the databases, 396 papers were retrieved from Scopus, 298 from IEEE Xplore, 63 from ScienceDirect, and 725 from Google Scholar. After removing duplicates and applying the inclusion/exclusion criteria, 121 papers remained [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], [75], [76], [77], [78], [79], [80], [81], [82], [83], [84], [85], [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97], [98], [99], [100], [101], [102], [103], [104], [105], [106], [107], [108], [109], [110], [111], [112], [113], [114], [115], [116], [117], [118], [119], [120], [121], [122], [123], [124], [125], [126], [127], [128], [129], [130], [131], [132], [133]. There has been a discernible upward trend in the publication of papers within the rPPG domain. The initial years saw a modest presence, but a notable surge in interest and research became evident, particularly peaking in 2023.

B. EXTRACTION AND SELECTION OF ALGORITHMS, DATASETS, AND EVALUATION SYSTEMS

From the selected articles, we extract benchmark algorithms used for comparison with their proposed method, the datasets used, and the evaluation metrics used. To identify the widely used benchmark methods, datasets, and evaluation metrics, we measured their utilization frequency in the selected articles, by counting the number of benchmark methods, datasets, and evaluation metrics. Based on the most highest frequency of the commonly picked number for each category, the widely used algorithms, public datasets, and evaluation metrics are chosen. Hence, we deliberately focused on the approaches most frequently adopted across the rPPG community, ensuring our framework captures a representative set of well-tested methods rather than relying on ad hoc or subjective selections. As illustrated in Figure 3, this frequency-based analysis guided us in pinpointing the

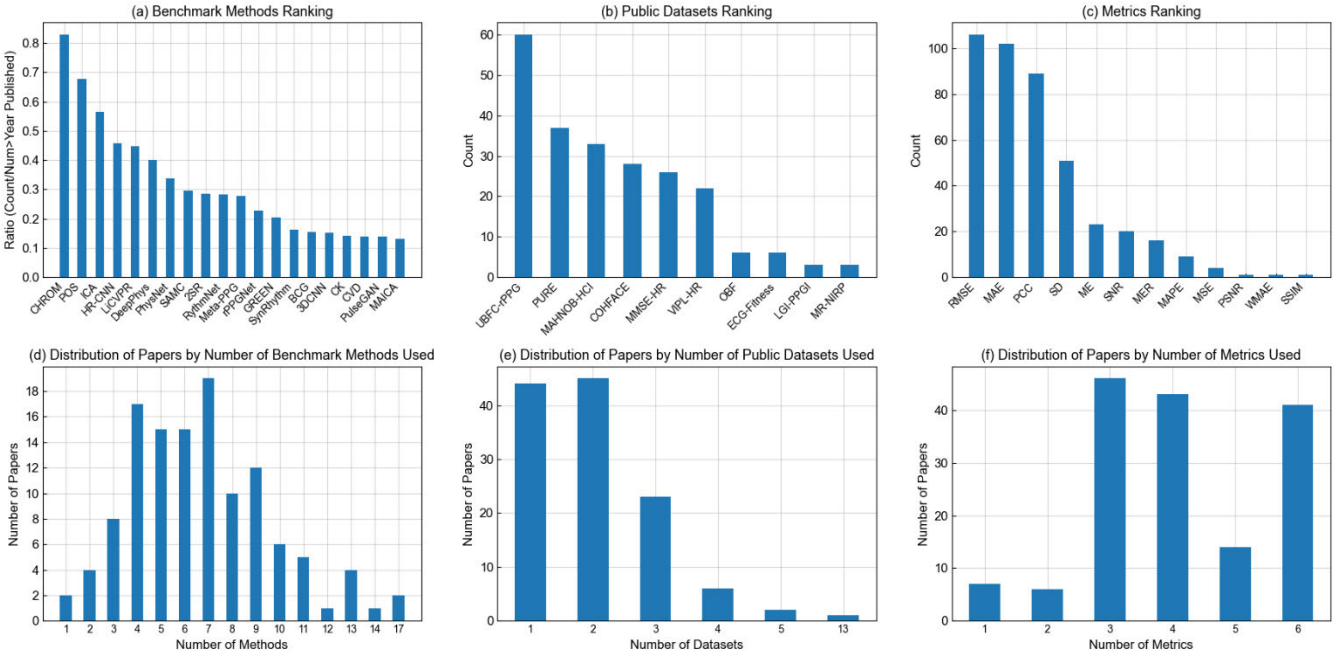


FIGURE 3. (a) Benchmark methods ranking, (b) Public datasets ranking, (c) Evaluation metrics ranking, (d) Histogram of benchmark methods used, € Histogram of the public datasets used, and (f) Histogram of the evaluation metrics used.

algorithms and datasets that consistently recur in the literature, thereby strengthening the generality of our framework.

Given the high discrepancy in the publication time of benchmark methods used, we calculate the total number of times the benchmark algorithm was used divided by number of publications since the introduction of benchmark algorithm ratio to account for the age of the published articles. If a paper use the same benchmark method across different datasets, it is only counted as once. To systematically demonstrate the high variability of accuracy reported for benchmark methods across public datasets, we also extract their reported results based on the evaluation metrics used on the applied datasets.

In line with common practice among unsupervised methods in this field, we focused on direct performance metrics (e.g., error measures against ground-truth heart rates) rather than advanced statistical testing procedures. As a result, most reviewed studies did not report in-depth statistical validation methods such as Bland–Altman plots or hypothesis testing, which further influenced our choice to emphasize straightforward performance measures. We also note that our selection naturally followed from an extensive literature review, wherein these three metrics (Figure 3c) consistently emerged as the de facto standard.

Furthermore, for the unsupervised methods, we dissect the proposed method into algorithms and assign them to specific stages (e.g., face detection and tracking, ROI extraction, filtering, signal normalization) based on its functionality and purpose. This information helps to identify the diverse approaches utilized in the multi-stage methods. We selected only the unsupervised methods because the majority of the selected supervised methods are end-to-end learning models, which means there are no pre-processing stages/algorithms and minimal to no post-processing stages/algorithms. Moreover, while deep-learning-based, end-to-end supervised methods hold promise, their monolithic architectures limit the ability to isolate, analyze, and refine individual preprocessing and postprocessing steps. Therefore, they could not be dissected and grouped the same way we did for unsupervised methods. Focusing on unsupervised methods thus provides a clearer pathway to systematically identify and optimize specific algorithmic components. For supervised methods, the training's nature – whether inter–dataset, intra–dataset, or a combination of both – was specifically identified. Moreover, for instances of intra-dataset training, the precise data splits used for training, validation, and testing were also recorded. We have provided a detailed description of the working principles of the widely used benchmark methods, the algorithms used and their parameters, selected public datasets, and evaluation metrics in the results section (Section III-B, Section III-C, and Section III-D respectively).

C. OPTIMAL ARCHITECTURE DESIGN

To identify the stages and associated algorithms in our proposed framework, we solely focus on established benchmark methods, as their substantial rate of utilization evidences their superior performance. We performed a similar process as the previous step to dissect the benchmark methods into distinct pre- and post-processing stages and captured the utilized algorithms. To identify the optimal architecture of algorithms for each core rPPG algorithm, the available algorithms at each

pre- and post-processing stages are permuted across selected datasets using the selected evaluation metrics. Specifically, we systematically iterate over each possible combination of these algorithms, evaluating each permutation's heart rate estimates against the ground truth and recording the resulting performance metrics. While these permutations could become highly computationally expensive, we demonstrate our results via an exhaustive search algorithm as we used high performance computing facilities available at the National University of Singapore. However, our proposed framework could utilize other types of search algorithms such as genetics algorithm and particle swarm optimization to alleviate the computational expense. This systematic approach ensures that all viable combinations are considered, allowing us to pinpoint the best-performing pipeline for each core rPPG method. To validate the performance of our proposed framework, we compare the most widely used accuracy metrics of the best performing architecture against the accuracies reported in the literature and the original implementations of the widely used rPPG methods. The proposed framework is illustrated in Figure 2.

The proposed framework begins by detecting a Region of Interest (ROI) from the input video frames, selecting areas like the forehead or lower face where subtle color changes related to blood flow are most detectable. Based on the literature, there are several robust algorithms used (e.g., Viola-Jones or MediaPipe based landmark tracking). Next, a pixel-filtering step excludes noisy or irrelevant data within this ROI, through RGB thresholding. This process removes background regions or areas affected by specular reflection, leaving primarily skin pixels. After pixel filtering, spatial averaging is applied across the cleaned pixels to produce a single raw signal representing color intensity changes over time. This raw signal is then segmented into overlapping windows (e.g., 1-second stride with a 5-second window size) to maintain temporal continuity. Within each window, one or more pre-filtering techniques may be employed such as bandpass filter (commonly 0.7-4 Hz to encompass typical heart rate ranges), moving-average filters for smoothing, or detrending filters for baseline removal, which are all designed to reduce motion artifacts and emphasize the pulsatile component in the signal. The refined signal is then processed by a core rPPG algorithm (e.g., CHROM, POS, ICA, LiCVPR, GREEN) to extract the heart rate waveform. Finally, post-filtering methods further reduce residual noise, and frequency or time-frequency analysis (e.g., FFT, PSD, STFT) is applied to estimate the heart rate accurately.

We adhered to the parameter choices as specified by each original benchmark method to maintain fidelity to their published designs. When certain parameters were not clearly stated or were ambiguous, we selected values based on common practices collected from our comprehensive literature review. This approach allowed us to replicate the typical usage scenarios described in the original papers rather than exhaustively optimizing every parameter, which lies outside the scope of this study. Future investigations could explore a broader parameter space to further tune each method for specific applications.

III. RESULTS

A. EXTRACTION AND SELECTION OF ALGORITHMS, DATASETS, AND EVALUATION SYSTEMS

1) WIDELY USED METHODS, PUBLIC DATASETS, AND EVALUATION METRICS

Figure 3(a) shows the top 20 most-widely used methods that the rPPG researchers use to benchmark their proposed methods. Figure 3(b) shows the top 10 most used public datasets to test proposed and benchmark methods and Figure 3(c) shows all the evaluation metrics they have used. The results were then ranked from highest to lowest.

The second row of Figure 3 shows the frequency of the total number of benchmark methods, public datasets and evaluation metrics used from all the papers. The bar chart shows that most papers tend to use 7 benchmark rPPG methods, 2 public datasets and 3 evaluation metrics. Supported by these findings, the top seven widely used algorithms were CHROM, POS, ICA, HR-CNN, LiCVPR, DeepPhys, and PhysNet. Given HR-CNN, DeepPhys, and PhysNet are end-to-end supervised methods, we could not dissect and include the algorithms as part of our proposed framework. Instead, we included Green as it is amongst the earliest rPPG implementations. The public datasets selected are UBFC-rPPG, PURE, and COHFACE. MAHNOB-HCI was not accessible, therefore was excluded. Similarly, the evaluation metrics chosen for this framework for accuracy assessments are MAE, RMSE and PCC.

2) BENCHMARK METHODS PERFORMANCE ACROSS PUBLIC DATASETS

Figure 4 illustrates the variability in the reported MAE results of same eight widely used benchmark rPPG methods across 10 public datasets. The box plots presented in Fig. 4 reveals significant variability in the MAE results reported which was extracted from the literature. The total number of papers, labelled as 'n', that reported the result for each public dataset at each core rPPG method is also indicated. This inconsistency primarily stems from the lack of a standardized baseline for the methods for the specific datasets to which they are applied, emphasizing the critical need for establishing one. Furthermore, certain studies report outliers — instances of anomalously high MAE values — that may skew the interpretation of their findings. This situation raises concerns that some studies may intentionally report worsened performance results for existing methods to exaggerate the comparative effectiveness of their proposed methods. The establishment of a robust and universally recognized baseline is imperative not only to ensure fair and accurate comparisons but also to uphold the integrity of research in the field of rPPG. Establishing such a benchmark would not only facilitate a more accurate comparison across different studies but also

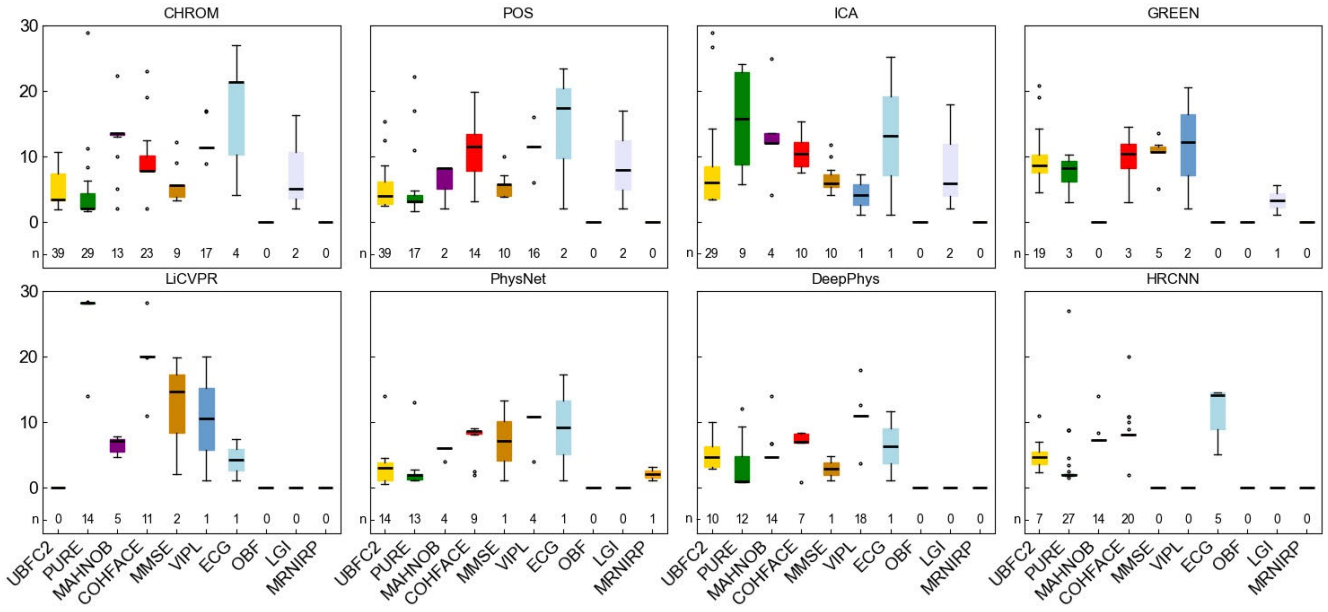


FIGURE 4. Reported MAE variability across public datasets for selected benchmark methods.

promote methodological rigor and potentially spur further innovation in the development of rPPG methods.

MAE has been chosen to illustrate the variability in Figure 4 as it provides a straightforward and transparent measure of average prediction error. Unlike RMSE, which can disproportionately penalize larger errors and amplify the effect of outliers, MAE treats all errors equally, offering a more balanced view of model performance. Furthermore, MAE is more interpretable than PCC since it quantifies the actual average deviation in heart rate estimates, rather than just the strength of a linear relationship.

3) DISSECTION OF THE STAGES OF ALL UNSUPERVISED ALGORITHMS

Table 1 and Table 2 provide a detailed summary of all the pre- and post-processing stages extracted from the unsupervised rPPG literature, as utilized by the researchers in their proposed remote heart rate detection systems. As it can be seen in Table 1, face detection and tracking, Region of Interest (ROI) extraction, normalization, filtering, and selection of color channels are among the stages with highest variability in the algorithms used, as these stages constitute the core pre-processes in rPPG.

As it can be seen in Table 2, the primary focus of post-processing is on eliminating noise from the PPG signal obtained during the rPPG signal extraction stage, and subsequently extracting the heart rate from this refined signal.

It can be observed that the majority of the studies place significant emphasis on bandpass filtering and transforming the signal into the frequency domain to obtain the frequency with the greatest amplitude.

However, this does not imply that a multitude of pre- and post-processing stages are necessary when proposing a new heart rate detection system. The purpose is to inform future researchers of the various options that have been explored, which may or may not be beneficial in their exploration phase. As demonstrated in this paper, certain pre- and post-processing algorithms are particularly effective with specific rPPG extraction methods. However, it is important to note that these algorithms are not universally scalable or effective with all datasets. In some cases, omitting certain stages might yield better results, highlighting the importance of tailoring the approach to the specific characteristics of the application scenario and dataset. Our proposed framework has been created with flexibility in such a way that the algorithms highlighted in the two tables above can be easily integrated and tested.

4) TRAINING AND TESTING DATA SEGMENTATION

Going a little deeper into the data extracted from the literature, Figure 5(a) presents a comparison of the number of unsupervised learning papers to supervised learning papers. At present, supervised learning algorithms have been proposed 2.5 times more frequently than unsupervised learning algorithms. A closer examination of these supervised learning papers, Figure 5(b), reveals that 38 of them focus on intra-dataset training (training and testing algorithms on different subsets of the same dataset), 10 on inter-dataset training (training on one dataset and testing on an entirely different dataset), 30 investigate both methods to assess the accuracy of their proposed algorithms, while 9 of the papers did not specify the way they trained and tested the algorithms.

For intra-dataset training, datasets can be split in various ways. The bar chart in Figure 5(c) provides specific details on these splits. In the x-axis, a notation a:b:c indicates that a% is used for training, b% for validation, and c% for testing.

TABLE 1. Number of unsupervised pre-processing stages and their respective algorithms.

Color Magnification		Face Detection and Tracking		ROI Extraction		Color Space Conversion		Normalize		Filtering		Selected Color Channel		Signal Decomposition		Head Pose Estimation		Signal Selection	
Eulerian Video Magnification	2	Viola Jones	16	ROI Patches	10	CHROM	4	Normalize	13	Detrending	10	RGB	14	EEMD	3	3D Head Pose Estimation	2	SNR	3
		KLT	10	AAM	2	HSV	3			Filtering	8	G	8	SVD	1			Modified Amplitude Selective Filtering	1
		DRMF	4	RGB-H Threshold	2	YCbCr	3					V	1	Wavelet Decomposition	1			PDS+IQR	1
		MTCNN	3	Ybor	2	POS	2					H	1						
		Dlib	2	Detected Face as ROI	2	CIE XYZ	1					Cr	1						
		MediaPipe	1	Viola Jones	2	LAB	1					A	1						
		InsightFace	1	Dlib	1	YIQ	1					Q	1						
		SeetaFace	1	MediaPipe	1							RGB-NIR	1						
		LibFaceDetection	1	ERT	1							R	1						
		Kernal Correlation Filters	1	Blue Red Difference Threshold	1														
		Custom	1	HSV	1														
		Intraface	1	RGB Threshold	1														
				Superpixel Segmentation	1														

KLT – Kanade-Lucas-Tomasi, DRMF – Discriminative Response Map Fitting, MTCNN – Multi-Task Cascaded Convolutional Neural Network, ROI – Region of Interest, AAM – Active Appearance Model, RGB-H – Red-Green-Blue-Height, ERT – Electrical Resistivity Tomography, HSV – Hue-Saturation-Value, CHROM – Chrominance, POS – Plane-Orthogonal-to-Skin, NIR – Near-Infrared, EEMD – Ensemble Empirical Mode Decomposition, SVD – Singular Value Decomposition, SNR – Signal to Noise Ratio, PSD+IQR – Power Spectral Density + Interquartile Range

TABLE 2. Number of unsupervised post-processing stages and their respective algorithms.

Filtering		Normalize		Signal Processing		HR Estimation		Outlier Removal	
Bandpass	12	Normalize	1	Overlap Adding	2	FFT	14	Outlier Removal	1
Detrending	3			EEMD	1	PSD	8		
Kalman Filter	2					Peak Detection	5		
NLMS	1					STFT	4		
Moving Average	1					DFT	1		

NLMS – Normalized Least Mean Squares Filter, FFT – Fast Fourier Transform, PSD – Power Spectral Density, STFT – Short Time Fourier Transform, DFT – Discrete Fourier Transform

The most used train-test split is 80% for training and 20% for testing. In one of the studies, the researchers trained their model on 50% of the dataset and subsequently tested it on the entire dataset, which included 50% of the data that the model had not previously encountered.

B. SELECTED ORIGINAL IMPLEMENTATIONS OF RPPG METHODS

Table 3 contains the different stages and its algorithms of the original implementation of the five most-widely unsupervised benchmark rPPG methods. The stages and algorithms in these five methods are integrated in our proposed framework. To further explain the rationale behind the design of each method, we have provided a brief description of the original implementation of the eight widely used benchmark methods below. This section is deliberately descriptive to capture each original implementation with maximum fidelity, ensuring our re-creation closely mirrors the methods as initially published.

1) CHROM

The CHROM method by [135] targets motion artifacts in non-contact PPG by filtering out specular reflection that lacks cardiac cycle information. The average RGB signals from skin regions are obtained by first detecting the face using Viola-Jones algorithm and applying a simple skin selection process using RGB thresholds. A sliding window of 1.6 seconds with a 0.8-second stride processes the average RGB signal, normalizing raw color channels by their means. Two orthogonal chrominance signals are computed as described in Equation 1 and Equation 2.

$$X_C = 3\bar{R} - 2\bar{G} \tag{1}$$

$$Y_C = 1.5\bar{R} + \bar{G} - 1.5\bar{B} \tag{2}$$

The pulse signal is derived by subtracting α-scaled Y_C and X_C, with α being the ratio of their standard deviations after bandpass filtering (0.67-4Hz or 40-240bpm). A Hanning window is applied to each signal segment before using overlap-adding with a 50% overlap to construct the final pulse signal. Heart rate is deduced by applying a sliding Fourier transform to the pulse signal over 12 seconds, incremented by 1 second, and selecting the frequency with the highest peak as the heart rate.

2) POS

The POS method by [136], like CHROM, aims to eliminate the reflection component in non-contact PPG, using in-phase signals rather than CHROM's anti-phase ones. While the original method prescribed by [136] uses OC-SVM for skin detection, due to lack of detailed parameters, this implementation follows a simpler skin detection process based on VJ face detection as used in CHROM above. From the windowed average RGB signals, with a 1.6-second window and one-frame stride, the plane orthogonal to skin signals X_p and Y_p are calculated as described in Equation 3 and Equation 4.

$$X_P = \bar{G} - \bar{B} \tag{3}$$

$$Y_P = \bar{G} + \bar{B} - 2\bar{R} \tag{4}$$

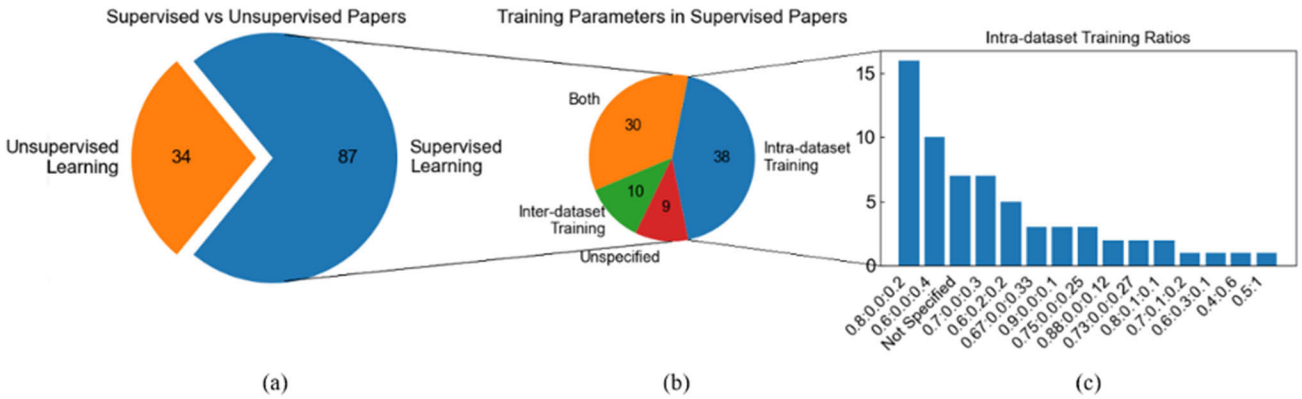


FIGURE 5. (a) Distribution of number of supervised and unsupervised papers, (b) Training parameters in supervised papers, (c) Frequency of intra-dataset training methods.

TABLE 3. Algorithms in pre- and post- processing stages of the top five benchmark methods.

	Pre-Processing Stages			Core rPPG Extraction Algorithm				Post-Processing Stages				
	Face Detection and Tracking	RGB Thresholding	Pre-Filtering					Post-Filtering			HR Estimation	Outlier Removal
CHROM	Viola Jones Bounding Box (Width: 1, Height: 1)	Yes	None	Signal Windowing	Normalize	Chrominance Signals	Overlap Adding	None			STFT (Continuous)	No
POS						Plane-to-Orthogonal Signals						
ICA	Viola Jones Bounding Box (Width: 0.6, Height: 1)	No				ICA Signal Decomposition Using JADE Algorithm					FFT (Continuous)	Yes
LiCVPR	Lower Face			Illumination Rectification		Non-Rigid Motion Elimination		Detrending	Moving Average	FIR Bandpass Filter	Welch's (Average and Continuous)	Yes
GREEN	1. Forehead Bounding Box 2. Pixel in Forehead 3. Region Beside Head 4. Whole Frame		1. None 2. Normalize 3. Butterworth Bandpass Filter	Green Channel				None			FFT (Average	

The pulse signal is derived by adding α-scaled Y_P and X_P, with α being the ratio of their standard deviations (without bandpass filtering). Overlap-adding, with a length of l-1 where l is the window length, normalizes S by subtracting its mean to derive the final pulse signal.

The heart rate is determined following a similar process to that in CHROM.

3) ICA

The ICA method by [137] is an early method for non-contact heart rate detection using a standard RGB camera. Faces are located in each frame using the Viola-Jones algorithm. The RGB signals are averaged from the pixels within the face ROI, which is narrowed to 60% of its original width while keeping the full height. The signals are windowed with a 30-second length and 1-second stride and normalized to zero-mean and unit variance. The JADE algorithm decomposes these normalized signals into three independent components. The second component, often reflecting the strongest PPG signal, is selected. Its power spectrum is obtained via FFT, identifying the peak frequency within the 0.75-4 Hz range (45-240 bpm) as the pulse frequency. An outlier removal step is added to exclude abrupt changes exceeding 12 bpm between consecutive seconds, opting for the next dominant frequency, or retaining the current rate if no alternative is found within the acceptable range.

4) HR-CNN

The HR-CNN method [131] is designed to counteract effects from illumination and subject motion for heart rate detection, utilizing a two-step CNN (an extractor and an HR estimator) that operates end-to-end without signal processing. During training, the extractor uses face images resized to 192×168 (aspect ratio 3:2) from the VJ detector, augmented through brightness adjustment, cropping, rotation, and translation. These frames and their corresponding ground truth heart rates, scaled by a factor of 60, train the extractor with the Adam optimizer at a learning rate of 0.0001. The extractor's output then trains the estimator, which uses the actual heart rate values with the same optimizer but a higher learning rate

of 0.1. For testing, the trained models are applied without augmentation. The extractor feeds into the estimator to predict the heart rate, and performance is assessed by comparing the estimated heart rate with the ground truth using Mean Absolute Error (MAE).

5) LICVPR

The LiCVPR method [100] addresses the challenge of environmental illumination and subject movement in heart rate detection. It refines initial face detection with DRMF to identify 66 facial landmarks and employs the KLT algorithm to track these landmarks accurately, even with head movement. The ROI is carefully selected from the lower face, excluding the eyes and edges to prevent non-facial pixel interference, focusing on the green channel. To counteract illumination variation, the mean green value of the ROI, influenced by blood volume and lighting, is corrected. Background mean green values provide a reference for environmental light effects, assuming similar illumination for the face and background in typical HCI settings. Using the Lambertian model assumption, a linear function estimates the correlation between face and background illumination changes. The Normalized Least Mean Square (NLMS) adaptive filter iteratively adjusts to minimize illumination errors, yielding an illumination-corrected signal.

For non-rigid motion noise, the signal is segmented and the standard deviation (SD) for each segment is computed. The segments with the highest SD, in the top 5%, are considered outliers and are removed since they are due to head movements. The segments are then reconstructed before applying the detrending filter, moving-average filter, and FIR Hamming window-based bandpass filter with a cut-off frequency of 0.7-4Hz (42-240bpm) to obtain the pulse signal. Finally, the heart rate is extracted from the filtered signal's power spectral density, calculated using Welch's method, by identifying the frequency with the highest power. Continuous heart rate estimation consists of detecting heart rate for each 10 second segments but average heart rate estimation consists of detecting a single average heart rate for the entire duration of the video. The codebase utilizes a continuous heart rate detection.

6) DEEPPHYS

The DeepPhys method [45] is an end-to-end system using a deep convolutional network for heart rate and breathing rate measurement, designed to work with head movements and variable lighting. The algorithm consists of a motion representation component and an attention mechanism. A subject-independent 5-fold cross-validation is used for training and testing. Faces are detected via the VJ algorithm, resized to 36×36, and inputted along with labels into the model. For motion representation, the normalized differences between consecutive frames are calculated and then clipped to three standard deviations. Meanwhile, the attention mechanism processes frames normalized to zero mean and unit standard deviation. Labels, which represent the first-order derivative calculated from differences between consecutive points in the PPG signal, are interpolated to align with the video's frame rate and then normalized. For testing, the motion and attention data feed into the model, and the resulting rPPG signal, along with ground truth PPG, is filtered using a 6th order Butterworth filter with 0.7-2.5Hz cutoff. Heart rate is then computed from the windowed signal using FFT. The model trains with a batch size of 128 for 48 epochs using the Adadelta loss function and a learning rate of 1e-4, as the original literature does not specify a rate.

7) PHYSNET

The PhysNet rPPG method [102] is the first end-to-end model that derives rPPG signals from videos without pre-processing, targeting heart rate (HR) and heart rate variability (HRV) detection. The literature divides spatio-temporal approaches into 3D Convolutional Neural Networks (3DCNN), which capture spatial and temporal contexts concurrently, and Recurrent Neural Networks (RNN), including LSTM variants, which emphasize temporal patterns within spatial features. Comparisons were made between PhysNet variations using 3DCNN and RNN frameworks and different loss functions: Mean Square Error (MSE) and Negative Pearson. The models underwent a subject-independent 10-fold cross-validation on public datasets, with face detection via the VJ algorithm, resizing to 128×128, and ground truth PPG signal resampling to video frame rate. Training used a 1e-4 learning rate, Adam Optimizer, and lasted 15 epochs.

Testing the trained model involved comparing the model's rPPG output with filtered and normalized ground truth PPG signals to determine heart rate through peak detection. The chosen parameters included a 4th order Butterworth bandpass filter (0.67-2Hz) and peak detection by FFT. The original research found optimal results with 128-frame clip lengths and the PhysNet-3DCNN-ED model with Negative Pearson loss. However, this paper's proposed framework performed best with 256-frame training length, which was consequently adopted.

8) GREEN

The GREEN method [138] demonstrates capturing PPG signals from the human face using ambient lighting and a basic video camera from several meters away. It extracts four ROIs from video frames: the forehead, a single pixel within the forehead, a non-skin area beside the head, and a downscaled full frame. The first two ROIs were obtained in the proposed framework using MediaPipe, and the third uses the VJ face detector. The green channel is averaged across ROI pixels to get the raw signal. The paper outlines three methods to compute heart rate from these ROIs using FFT peak frequency analysis. The first uses the unaltered green signal; the second normalizes by subtracting the mean; the third applies a Butterworth bandpass filter, either 0.8-2Hz or 0.8-6.0Hz. The most effective approach found in the framework is the forehead ROI with a 0.8-2Hz bandpass filter.

C. SELECTED DATASETS

In our framework proposed in this paper, the ground truth PPG signals were processed in a similar way as the reconstructed signals obtained by the selected rPPG methods, unless mentioned otherwise in the original literature. Here we describe the three most widely used datasets in the literature applied in evaluating the performance of our proposed framework.

1) UBFC-RPPG DATASET

The UBFC dataset [139] includes 50 videos, with UBFC1 comprising 8 and UBFC2 containing 42. Recorded at 30 fps and a 640 × 480 resolution using the Logitech C920 HD Pro, UBFC1 has about 16,500 frames, while UBFC2 features minute-long videos. Subjects were filmed approximately 1 meter from the camera in various levels of sunlight and indoor lighting conditions. The reference PPG signal was captured at 60Hz by a CMS50E pulse oximeter. For UBFC2, the PPG signal is downsampled to 30Hz, whereas UBFC1 maintains the original 60Hz. Although both subsets are included in the proposed framework, results are reported only for UBFC2, the subset more commonly used for rPPG studies.

2) PURE

The PURE dataset [140] features 60 videos of 10 participants executing 6 head movements: steady, talking, slow and fast translations, and small and medium rotations, each video lasting roughly a minute. Filmed with an eco274CVGE camera at 30 fps in 640 × 480 resolution, the dataset pairs with PPG signals from a CMS50E pulse oximeter recorded at 60Hz. Participants were positioned 1.1 meters from the camera.

For steady motion, subjects stayed still; during talking, they spoke without moving their heads. In translation motions, subjects followed a rectangle moving at 7 pixels per second for slow and 14 pixels per second for fast. Rotation motions required head turns towards targets 35 cm away for a 20° small rotation and 70 cm away for a 35° medium rotation.

3) COHFACE

The COHFACE dataset [8] features 160 videos from 40 subjects under two lighting conditions: studio and natural. Studio lighting used closed blinds and artificial light, while natural lighting had open blinds with no extra illumination. Videos were captured with a Logitech HD C525 camera at 20 fps and 640 × 480 resolution. Reference PPG signals were recorded at 256 Hz using a SA9308M BVP sensor, with each video lasting about 60 seconds. Note that in multiple literature papers, COHFACE is specified to be a lower quality dataset due to how the videos were compressed.

D. SELECTED EVALUATION METRICS

The evaluation metrics – RMSE, MAE, and PCC – have been selected to assess the accuracy of the estimated heart rate (HR_{ES}) in comparison to the ground truth heart rate (HR_{GT}). This was done for each dataset and for both the original and improved versions of the rPPG algorithms. Benchmark methods often produce a continuous HR_ES and HR_GT for a single video due to the incorporation of signal windowing. As such, before these evaluation metrics are applied, the values are averaged.

1) ROOT MEAN SQUARED ERROR (RMSE)

The Root Mean Squared Error (RMSE) measures the average difference between HR_{ES} and HR_{GT} by determining the square root of the mean squared differences as described in Equation 5.

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(HR_{GT_i} - HR_{ES_i}\right)^2} \tag{5}$$

It provides an estimate of the magnitude of these differences, with higher values indicating larger discrepancies. In essence, RMSE represents the standard deviation of the errors in heart rate estimation, with smaller RMSE values indicating a better estimated heart rate.

2) MEAN ABSOLUTE ERROR (MAE)

The Mean Absolute Error (MAE) measures the error between HR_{ES} and HR_{GT} by determining the mean of the absolute differences as show in Equation 6.

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|HR_{GT_i} - HR_{ES_i}\right| \tag{6}$$

3) PEARSON CORRELATION COEFFICIENT (PCC)

The Pearson Correlation Coefficient (PCC) determines the correlation between HR_{ES} and HR_{GT} as demonstrated in Equation 7.

$$PCC = \frac{\sum_{i=1}^{N}\left(HR_{GT_i} - \mu_{GT_i}\right)\left(HR_{ES_i} - \mu_{ES_i}\right)}{\sqrt{\sum_{i=1}^{N}\left(HR_{GT_i} - \mu_{GT_i}\right)^2}\sqrt{\sum_{i=1}^{N}\left(HR_{ES_i} - \mu_{ES_i}\right)^2}} \tag{7}$$

E. OPTIMAL ARCHITECTURE DESIGN RESULTS FOR SELECTED PUBLIC DATASETS

The boxplots in Fig. 6 illustrate the outcomes derived from both the original implementations and the best combination of algorithms drawn from permutating through different pre- and post-processing algorithms for different stages (Table 3) via our proposed framework. Detailed and specific numerical results are available in Table 4, along with the top performing algorithm for each pre- and post-processing stages for the specific core rPPG algorithm. Furthermore, owing to the distinctive characteristics of each dataset, including variations in video quality and environmental recording conditions, different optimal algorithms emerge for each stage, as indicated in Table 4.

Notably, 80% of the enhanced outcomes achieved through the flexible framework fall within the 25th percentile of

TABLE 4. Algorithms in pre- and post- processing stages of the top five benchmark methods.

	Pre-Processing Stages				Core rPPG Extraction Algorithm	Post-Processing Stages				Results					
	Face Detection and Tracking	RGB Thresholding	Pre-Filtering			Post-Filtering		HR Estimation	Outlier Removal	MAE		RMSE		PCC	
										Original	frPPG	Original	frPPG	Original	frPPG
UBFC2	Forehead Bounding Box	Yes	None		CHROM	Butterworth Bandpass Filter		STFT (Continuous)	No	1.86	0.66	3.41	1.07	0.98	1.00
					POS	Moving Average				5.35	0.75	9.21	1.88	0.86	0.99
	Lower Face	No	Detrending		ICA	Detrending	FIR Bandpass Filter	Welch's (Continuous)	Yes	14.1	7.88	22.24	14.93	0.48	0.71
			Detrending	FIR Bandpass Filter	LiCVPR				No	15.26	7.07	26.11	13.45	0.33	0.69
					GREEN	Detrending	Moving Average	FFT (Average)	Yes	7.11	2.31	13.07	4.69	0.73	0.96
PURE	Viola Jones Bounding Box (Width: 0.6, Height, 1)	Yes	Detrending		CHROM	Moving Average		FFT (Continuous)	No	3.77	2.54	8.09	3.29	0.94	0.96
		No	None		POS	Detrending	Moving Average			2.39	1.48	4.7	2.58	0.98	0.99
			Detrending		ICA	Butterworth Bandpass Filter		STFT (Continuous)	Yes	11.46	5.77	21.66	11.25	0.51	0.88
		Yes	Detrending	Butterworth Bandpass Filter	LiCVPR	Detrending	Butterworth Bandpass Filter	Welch's (Continuous)	No	7.22	5.25	16.32	11.77	0.48	0.74
	Lower Face		Butterworth Bandpass Filter		GREEN			FFT (Average)	Yes	9.71	3.53	19.82	8.49	0.53	0.94
COHFACE	Viola Jones Bounding Box (Width: 1, Height, 1)	Yes	Detrending		CHROM	Moving Average		Welch's (Continuous)	No	15.12	9.69	17.91	11.67	0.42	0.57
	Viola Jones Bounding Box (Width: 0.6, Height, 1)				POS	Moving Average	FIR Bandpass Filter	FFT (Continuous)		8.66	6.11	10.78	8.29	0.44	0.67
			Detrending	Moving Average	ICA	Detrending	Butterworth Bandpass Filter	STFT (Continuous)		16.71	8.24	28.73	13.63	0.51	0.88
			Moving Average	Butterworth Bandpass Filter	LiCVPR	Moving Average		FFT (Continuous)		8.03	6.4	12.97	10.91	0.48	0.74
	Forehead Bounding Box	No			GREEN	Moving Average	Butterworth Bandpass Filter	Welch's (Average)	Yes	9.93	7.48	14.13	10.6	0.33	0.51

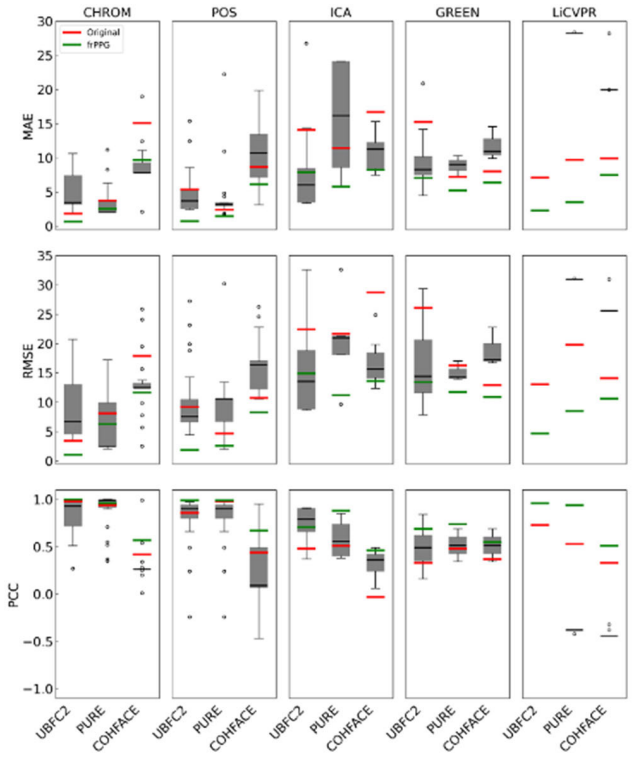


FIGURE 6. MAE, RMSE, and PCC results from our proposed framework (green lines), the original implementations (red lines), and the reported results in the literature (boxplots) of the benchmark rPPG methods across the three widely used public datasets.

reported MAE, RMSE, and PCC results, demonstrating significant variability. Additionally, 60% of the improved results surpass the lowest reported outcomes found in the literature. It should be also noted that the results that had been reported from the literature does not include the source code, hence it cannot often objectively evaluate their validity. This highlights the necessity of an open codebase where users can exactly replicate the results.

Considering the variability observed in the results across all core rPPG algorithms, we have obtained the lowest results compared to best the reported results for UBFC2 and PURE dataset. To be more specific, for UBFC2 dataset, CHROM frPPG achieved a MAE of 0.66, which is 66% lower than the lowest reported MAE in the literature which was 1.84. For the PURE dataset, our POS frPPG also achieved the lowest MAE of 1.48, where the lowest reported MAE was 1.62. However, as for the COHFACE dataset, as reported in literature papers [83], [142], the quality of the dataset is significantly lower as compared to the other datasets since the videos are heavily compressed.

F. CODEBASE

Our codebase contains the framework introduced in this paper as well as original implementations of the widely used rPPG methods. The flexible framework (frPPG) contains all algorithms shown in Table 3 and used in Section III-E. In the raw signal extraction stage, users have seven different regions of interest to select from, given that CHROM ROI and POS ROI is implemented in the same way as mentioned in Section III, complemented by an option for simple RGB thresholding to filter noisy pixel. Following this, there are three distinct pre-filtering methods. These methods can be either combined or executed independently. Notably, among the filtering methods is a bandpass filter, which comes in two variations: an FIR bandpass filter with a Hamming window and a Butterworth bandpass filter. As it would be counterproductive to layer these two filters, the total number of filter combinations stands at 12. Users can then proceed to

select a core rPPG extraction algorithm, followed by the post-filtering algorithms, using the same options available in pre-filtering. Finally, the heart rate is extracted using one of three methods: FFT, PSD, or STFT. These algorithms have been enhanced with an option to eliminate outliers; a feature drawn from the original ICA implementation. The user can permutate through the different algorithms in each stage to identify best architecture of the algorithms that suits the core PPG signal algorithm. In total, our codebase contains 12,096 possible predefined permutations for each core rPPG algorithm. This framework is meticulously designed to grant users comprehensive control over many critical variables. This encompasses aspects like window size, stride, cutoff frequencies, face ROI dimensions, and more. Such segregation in the codebase allows the users to evaluate the original versions of the selected rPPG algorithms or their refined versions, or refine their proposed rPPG method. The codebase also contains three evaluation metrics: MAE, RMSE, and PCC and is readily compatible with three major public datasets: UBFC-rPPG, PURE, and COHFACE.

IV. DISCUSSION

While the current implementation of our proposed framework (frPPG) contains the algorithms in pre- and post-processing stages of five widely used rPPG methods, it could also incorporate additional algorithms or new stages. This adaptability and modularity are conducive to continual improvement in remote heart rate estimation by making it easier to develop more complex rPPG methods. The computational cost for permutating through algorithms and their hyperparameters is another important factor in developing and evaluating novel rPPG methods. For example, there were approximately 12,096 permutations (algorithms and hyperparameters) for each core PPG algorithm in our implementation. In our high-performance computing facility, each permutation on a public dataset took about 30 seconds. Due to the computational expenses, we were unable to explore a more discretized space of the parameters, which might have further improved the accuracy results. While computational efficiency is indeed a practical consideration, a detailed complexity analysis lies outside the primary scope of this study. The algorithms employed remain relatively lightweight for typical public rPPG datasets; however, future large-scale or real-time applications may warrant a more thorough investigation of runtime performance. Our proposed framework demonstrated considerably low MAE results for the majority of the scenarios. This is because noises were attenuated in the pre- and post-processing stages. These benefits are intrinsically lacking in the end-to-end supervised rPPG models. Therefore, hybrid rPPG methods in which the raw signals are pre-processed before passing it through a deep learning model to train and estimate the rPPG signal could be explored in the future [131]. Our framework and codebase could support such future explorations.

Despite its merits, our current approach also has some limitations. First, the computational cost of exhaustive permutations can become prohibitively high for large-scale experiments or highly complex parameter spaces. Additionally, the framework currently focuses on a fixed set of widely used algorithms and does not automatically generalize to novel or domain-specific algorithms without manual integration efforts. Another limitation is that the framework predominantly addresses scenarios with relatively controlled conditions (e.g., standard public datasets) and may require adaptation or refinement when applied to more challenging, real-world environments. Furthermore, the end-to-end supervised methods, while discussed, remain less amenable to the modular optimization strategies employed here, thus highlighting a potential gap in the current design.

A necessary requirement for the success of any rPPG method is that the dataset (i.e., image series) contains adequate rhythmic color variability due to the heartbeats in the skin areas. The signal strength could be impacted by various factors such as variation in illumination, motion artifacts, distance from the camera, and quality of the imaging sensors [11]. Therefore, a signal quality indexing system is required to verify if the subject is in a suitable environment, with proper lighting, at an appropriate distance, and experiencing minimal head movements, and using a good-quality camera. For example, as the distance between the camera and the subject increases, the noises and heart rate frequency peaks mesh together in the frequency spectrum. The current available public datasets are often created in a controlled environment with the aim to minimize the environmental noise, which is not the case in real-time scenarios. One way to attenuate the illumination variations is by switching to near infrared rPPG detection, which can minimize the effects of visible light to be at a minimum [141].

V. CONCLUSION

rPPG has been gaining traction over the past few years, where novel methods and algorithms have been developed and proposed to improve heart rate estimation accuracy. However, the evaluations of rPPG methodologies are fragmented and is evident by the high variability of accuracy reported for benchmark methods across public datasets. This is primarily due to that there is an interdependency between the core rPPG methods, and the pre- and post-processing stages and researchers arbitrarily chooses the algorithms.

To address this issue, we conducted a comprehensive literature review for original articles with published after 2005. After screening the articles, we selected 121 papers that met our criteria by examining abstracts and, if necessary, the full documents. From these papers, we extracted key information, including benchmark algorithms, datasets, evaluation metrics, training and testing data segmentation, reported results, and algorithms used in pre- and post-processing stages. Identifying widely used rPPG methods, public datasets, and evaluation metrics was based on the most frequent occurrences in each category across all studies. Based on this, we have identified the widely used rPPG methods as CHROM, POS, ICA, HR-CNN, LiCVPR, DeepPhys,

PhysNet, and GREEN, public datasets as UBFC-rPPG, PURE and COHFACE, and evaluation metrics as RMSE, MAE and PCC. Subsequently, we illustrate the Mean Absolute Error (MAE) variability from reported results of benchmark rPPG methods on all datasets, underscoring the imperative for a benchmark in this field.

Based on the findings, we proposed a framework to optimally select the algorithms in pre- and post-processing stages, which was done by dissecting the original implementation of the five widely used unsupervised rPPG methods. Through permutating the algorithms at each step, we were able to improve the core rPPG extraction algorithm by selecting the optimal pre- and post-processing algorithms. Our results indicate that 80% of the refined methods fall within the 25th percentile of the reported MAE, RMES and PCC results. Moreover, 60% of the refined methods surpassed the lowest reported results across the literature. All the refined methods were better than the original implementations. We have made the codebase for this framework publicly accessible to enable users to either improve on their methods or to check and test existing methods. While the current implementation of our proposed framework (frPPG) contains the algorithms in pre- and post-processing stages of five widely-used rPPG methods, it could also incorporate additional algorithms or new stages.

In the future, this framework could be extended by incorporating automated algorithm selection procedures (e.g., genetic algorithms or reinforcement learning) to reduce computational overhead and enable more efficient navigation of the parameter space. Furthermore, integrating additional datasets with more diverse environmental conditions (e.g., varying lighting, movement, distance to the camera) could strengthen the generalizability of the findings. Exploring hybrid approaches, where preprocessed signals serve as inputs to deep learning models, may also bridge the gap between the interpretability of modular methods and the adaptive power of end-to-end architectures. These future directions have the potential to further reduce the variability in reported results and enhance the real-world applicability of rPPG methods.

ACKNOWLEDGMENT

Any opinions, findings, and conclusions expressed in this material are those of the authors and do not necessarily reflect the views of Singapore Ministry of Education.

CODE AVAILABILITY

Our codebase is available publicly in an online repository at https://github.com/Building-Robotics-Lab/flexible_rPPG. The three public datasets used can be requested for access from: UBFC-rPPG [139]: https://sites.google.com/view/ybenezeth/ubfcrppg, PURE [140]: https://www.tu-ilmenau.de/universitaet/fakultaeten/fakultaet-informatik-und-automatisierung/profil/institute-und-fachgebiete/institut-fuer-technische-informatik-und-ingenieurinformatik/fachgebiet-neuroinformatik-und-kognitive-robotik/data-sets-code/pulse-rate-detection-dataset-pure, and COHFACE [8]: https://www.idiap.ch/en/scientific-research/data/cohface.

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